Concept Map 1: Decision Support System (DSS)

TYPES OF DSS

- Data-driven DSS
- Model-driven DSS
- Knowledge-driven DSS
- Communication-driven DSS

FLOW: DATA → ANALYSIS → DECISION



DECISION SUPPORT SYSTEM (DSS)

ROLE IN BUSINESS DECISION-MAKING

- Supports semi-structured & unstructured decisions
- Improves decision quality
- Reduces risk & uncertainty
- Helps in strategic planning

TYPES OF DATA USED

- Internal Data
- External Data
- Real-time Data
- Historical Data

CORE COMPONENTS

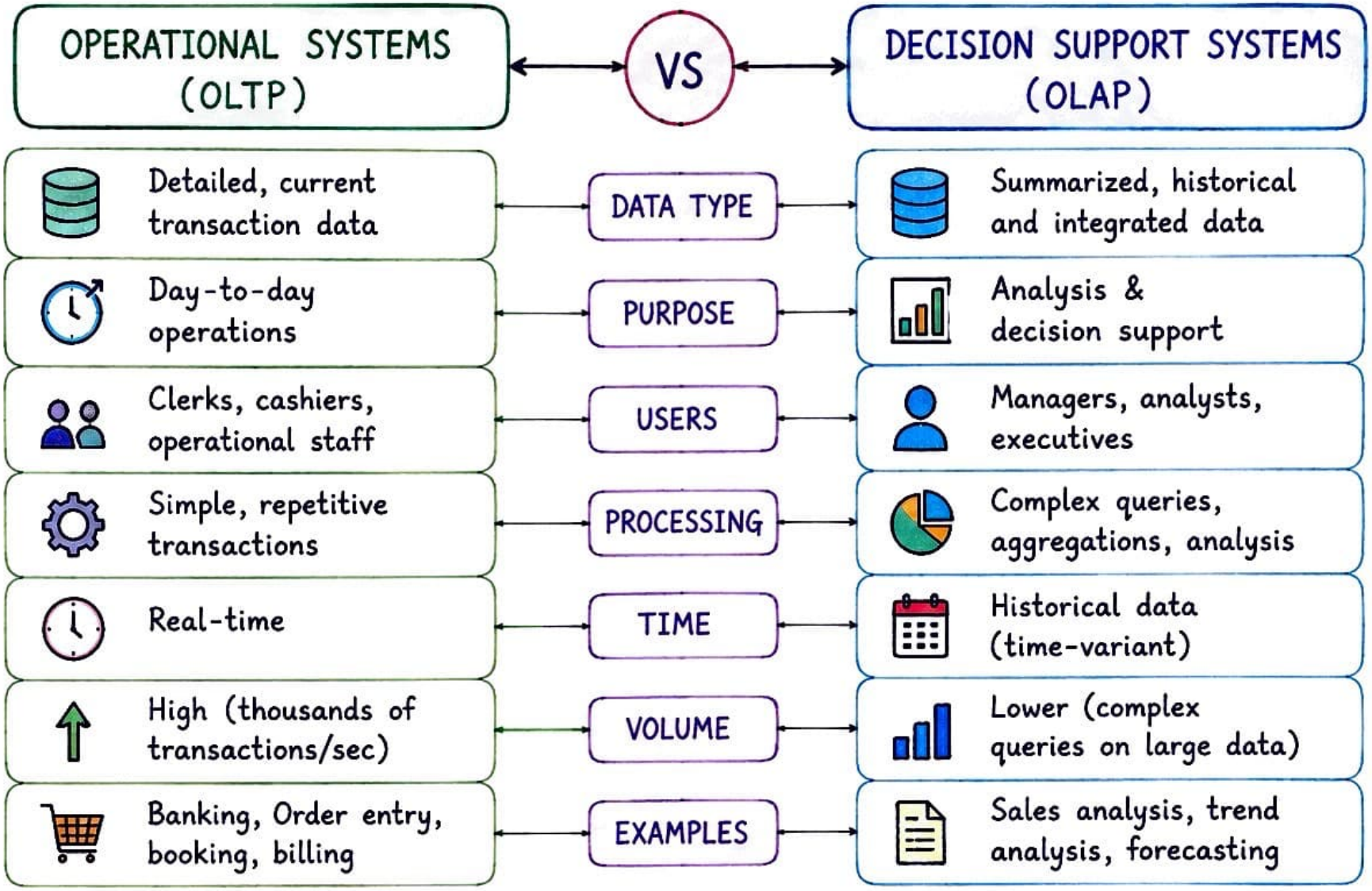
- Data Management
- Model Management
- User Interface
- Knowledge Management

REAL-WORLD EXAMPLE

Sales forecasting DSS helps a retail company decide inventory, discounts & promotions.

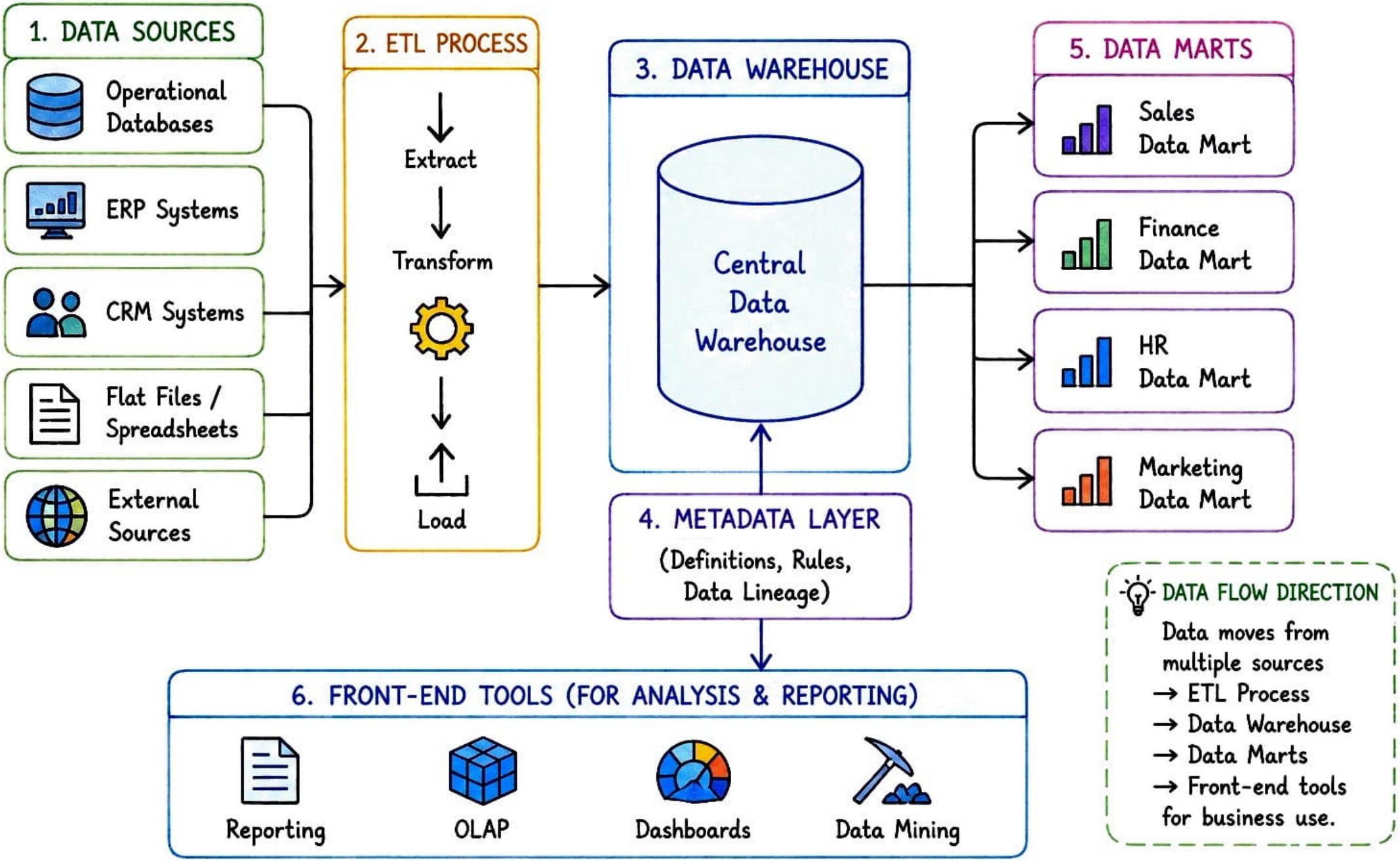
★ DSS helps organizations make better decisions by using data, models, knowledge and analytical tools. ★

Concept Map 2: Operational Systems vs DSS



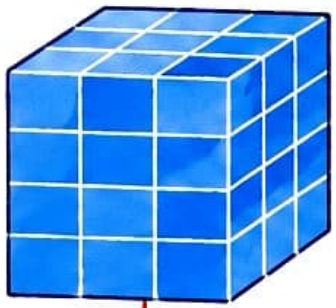
SIMILARITIES: Both use databases, ensure data integrity, and support business functions.

Concept Map 3: Architecture of Data Warehouse



Concept Map 4: OLAP, Data Cubes & Dimensional Modelling

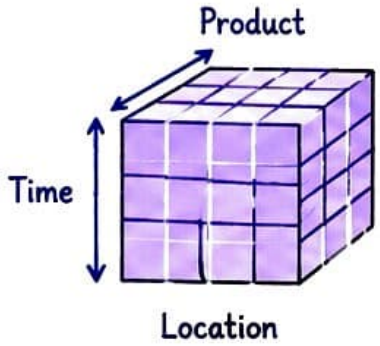
OLAP



1. OLAP OPERATIONS

- Slice (2D subset)
- Dice (Sub-cube)
- Roll-up (Summarize)
- Drill-down (Detailed view)
- Pivot (Rotate view)

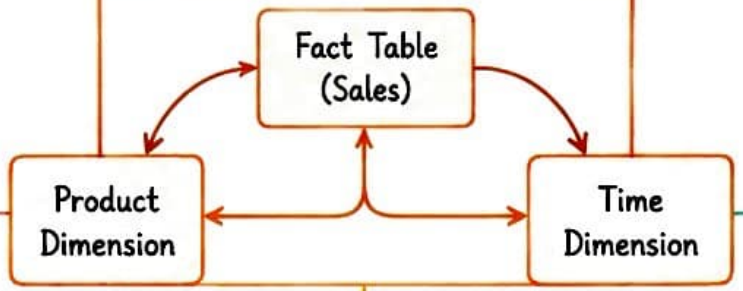
2. DATA CUBE STRUCTURE



Dimensions: Product, Time, Location

Measure: Sales Amount

3. STAR SCHEMA



4. DIMENSION TABLES

- Store descriptive data
- Low volume
- Example: Product, Time, Location

5. FACT TABLE

- Stores measurable data
- High volume
- Contains foreign keys to dimensions
- Example: Sales Amount, Quantity, Discount

6. ANALYTICAL USAGE

Trend analysis | Performance measurement | Forecasting | What-if analysis

LEGEND

→ Data / Flow

- - -→ Operation

Concept Map 5: Query Processing & BI Tools (KNIME)

1. QUERY PROCESSING STAGES

1. Parsing
- Check syntax & validate query

2. Optimization
- Find efficient execution plan

3. Execution
- Execute plan & retrieve data

2. ROLE OF BI TOOLS

- Connect to data sources
- Data preparation
- Analysis & visualization
- Reporting & dashboards
- Support decision-making

3. KNIME WORKFLOW STRUCTURE

Node Repository → Workflow Canvas → Execute Workflow → Results (View / Table / Chart)

4. DATA → QUERY → VISUALIZATION PIPELINE

Data (Input) → Query (Processing) → Visualization (Output)

5. OLAP APPLICATION INTEGRATION

Use OLAP data sources (Cubes) → Perform drill-down, roll-up, slice & dice operations → Visualize insights using BI tools like KNIME, Power BI, Tableau, etc. → Generate interactive reports & dashboards → Enable better analysis & decision-making